

USER INFORMATION
INJEC SOL K10
POTASSIUM CHLORIDE 10% w/v INJECTION BP
Sterile and Non-Pyrogenic Concentrate

Composition:

Each 10mL Solution Contains:

Potassium Ion (K ⁺)	13.4 mmol
Chloride Ion (Cl ⁻)	13.4 mmol

Product Description:

Potassium Chloride 10% w/v Injection BP is a sterile and non-pyrogenic concentrate solution in Water for Injections BP.

Pharmacology:

Pharmacodynamics:

Potassium is the major cation of intracellular fluid and is essential for maintenance of acid-base balance, isotonicity and the electrodynamic characteristics of the cell. Potassium ions are involved in a number of essential physiological processes, including the maintenance of intracellular tonicity, the transmission of nerve impulses, the contraction of cardiac, skeletal, and smooth muscle and the maintenance of normal renal function. Potassium chloride is used as a source of the potassium cation for treatment or prevention of potassium depletion in patients in whom dietary measures are inadequate. Potassium chloride may also be used cautiously to abolish arrhythmias or cardiac glycoside toxicity precipitated by a loss of potassium.

Pharmacokinetics:

Potassium chloride is generally readily absorbed from the gastro-intestinal tract. Potassium is excreted mainly by the kidneys; it is secreted in the distal tubules which are also the site of sodium-potassium exchange. After intravenous administration, potassium is actively transported from extracellular fluid into cells where concentrations reach up to 40 times that of extracellular fluid. The capacity of the kidneys to conserve potassium is poor and urinary excretion of potassium continues even when there is severe depletion. Tubular secretion of potassium is influenced by several factors, including chloride ion concentration, hydrogen ion exchange, acid-base equilibrium and adrenal hormones. Some potassium is excreted in the faeces and small amounts may also be excreted in saliva, sweat, bile and pancreatic juice.

Indication:

Indicated for the treatment of potassium depletion in patients with hypokalaemia. The intravenous route is indicated when the patient is unable to take potassium orally or if hypokalaemia is severe.

Recommended Dosage:

This product contains no antimicrobial preservative and should be used in one patient on one occasion only. INJEC SOL K10 is a concentrated solution and **MUST BE DILUTED BEFORE USE** by adding to a large volume intravenous fluid before use. For example, 10mL diluted with not less than 500mL 0.9% Sodium Chloride Intravenous Infusion BP, or other suitable diluent, and mixed well. Administration is by slow IV infusion. Too rapid infusion of hypertonic solutions may cause hyperkalaemia, local pain and rarely, vein irritation. Adjust rate of administration according to tolerance. Use of the largest peripheral vein and a small bore needle is recommended.

The dosage and rate of administration should be individualised based on patient's age, weight, clinical and biological (acid-base balance) conditions of the patient and concomitant therapy, ECG and serum electrolyte determinations.

THE USUAL MAXIMUM CONCENTRATION IS 40 mmol/L and usual maximum recommended dose of potassium is 2 to 3 mmol/kg/24h.

In patients whose serum potassium concentration is above 2.5 mmol/L, the rate of infusion should not exceed 10 mmol/hour. The total dose should not exceed 200 mmol/24 hours.

If urgent treatment is required (serum potassium concentration less than 2 mmol/L with ECG changes or paralysis), infuse potassium in a suitable concentration at a rate of 40 mmol/hour, up to a rate of 400 mmol/24-hour period.

In critical states, potassium may be infused in saline (unless saline is contraindicated) rather than in glucose solutions, as the latter may decrease serum potassium concentrations, producing an intracellular shift. To reduce microbiological hazard, use as soon as practical after dilution.

Note: 10mL of INJEC SOL K10 solution contains 1g of Potassium Chloride and 1g of Potassium Chloride provides 13.4mmol each of Potassium ion and Chloride ion.

Route of Administration : I.V. Infusion.

Contraindications:

- Hyperkalaemia
- Severe renal impairment with oliguria, anuria, or azotaemia,
- Hyperchloraemia,
- Acute ischaemic stroke,
- Head trauma (first 24 hours)
- Untreated Addison's disease
- Adynamia episodica hereditaria
- Acute dehydration
- Heat cramps
- Metabolic acidosis

Warnings and Precautions :

INJEC SOL K10 MUST NOT BE INJECTED UNDILUTED. DIRECT INJECTION WITHOUT APPROPRIATE DILUTION MAY CAUSE INSTANT DEATH. FOR SLOW I.V. INFUSION ONLY. NOT FOR I.V. BOLUS.

Potassium should be administered with considerable care to patients with cardiac disease or conditions predisposing to hyperkalaemia such as renal or adrenocortical insufficiency, acute dehydration, extensive tissue destruction as occurs with severe burns and concomitant use of a Potassium-sparing diuretic or ACE inhibitor regular monitoring of clinical status, serum electrolytes and ECG is advisable in patients receiving potassium therapy, particularly those with cardiac or renal impairment. Glucose can reduce serum-potassium concentrations, and glucose-free solutions should be used for the initial intravenous administration of potassium in hypokalaemia.

Caution:

The compatibility of the large volume IV fluid intended for dilution should be checked before use. Single dose container. Discard all unused contents. Do not use if leakage is detected. If any visible solid particles, growth or turbidity appears during storage, the product should not be used.

Interactions with Other Medicaments:

Potassium supplements should be used with caution, if at all, in patients receiving drugs that increase serum-potassium concentrations. These include potassium-sparing diuretics, ACE inhibitors, cyclosporine, and drugs that contain potassium such as the potassium salts of penicillin. Similarly, the concomitant use of potassium-containing salt substitutes for flavouring food should be avoided. Beta-adrenergic blockade increases both peak serum potassium concentration and the time required for serum potassium to return to basal levels in subjects receiving an acute intravenous potassium load. Nonsteroidal anti-inflammatory drugs (NSAIDs) may cause hyperkalaemia by inducing secondary hypoadosteronism following inhibition of renal prostaglandin synthesis.

Heparin reduces the synthesis of aldosterone which may result in hyperkalaemia, especially in patients with underlying renal insufficiency or other problems that impair potassium excretion.

Potassium supplementation is not recommended with concurrent use of digitalis glycosides in those patients with severe or complete heart block. Careful monitoring is necessary if potassium chloride is used to correct hypokalaemia in such patients.

Concurrent use with insulin may decrease serum potassium.

Concurrent use with sodium bicarbonate may decrease serum potassium.

Corticosteroids are associated with the retention of sodium and water, with oedema and hypertension.

Incompatibilities:

As guidance, the solutions are incompatible with the following medications:

- Diazepam
- Ergotamine
- Methylprednisolone
- Phenytoin
- Sodium Nitroprusside

Additives known to be incompatible should not be used.

Statement on usage during pregnancy and lactation:**Use in pregnancy:**

Both potassium and chloride ions are essential constituents of human tissues and fluids. However, supraphysiological levels of potassium are detrimental to maternal and fetal cardiac function. Serum levels should be closely monitored in pregnant women receiving potassium therapy.

Use in lactation:

Potassium is excreted into breast milk. Because of the potential for potassium to cause adverse effects on the breastfeeding baby, caution should be exercised when potassium therapy is given to a breastfeeding woman. Potassium doses that does not result in hyperkalemia for the mother is not likely to adversely affect the fetus or affect the potassium content in human breast milk. It may be used during pregnancy and lactation under the supervision of the prescribing physician.

Adverse Effects/Undesirable Effects:

Adverse reactions may be associated with the technique of administration, including febrile response, infection at the site of injection, local pain or reaction, vein irritation, venous thrombosis or phlebitis extending from the site of injection, extravasation, and hypervolemia.

Overdose and Treatment:

Excessive administration of potassium may lead to the development of hyperkalaemia, especially in patients with renal impairment. Symptoms include paresthesia of the extremities, muscle weakness, paralysis, cardiac arrhythmias, heart block, cardiac arrest, and mental confusion.

One of the important indicators of potassium toxicity is ECG change including tall, peaked T-waves, depression of S-T segment, disappearance of the P-wave, prolongation of the Q-T interval, and widening and slurring of the QRS complex.

In the event of accidental over infusion, treatment should be discontinued and the patient should be observed for the appropriate signs and symptoms related to the drug administered. The relevant symptomatic and supportive measures should be provided as necessary.

Treatment of hyperkalaemia involves the administration of calcium, insulin or sodium bicarbonate, and exchange resins or dialysis. Parenteral frusemide with substantial doses of sodium chloride and bland fluids will assist excretion of excess potassium through the kidneys.

Excessive administration of chloride salts may cause a loss of bicarbonate with an acidifying effect.

Storage Conditions:

Do not store above 30°C

Shelf Life:

3 years from manufacturing date in proper storage condition in LDPE Ampoule. Do not use after expiry.

Dosage forms and packaging available:

10mL X 20 LDPE plastic ampoules per box

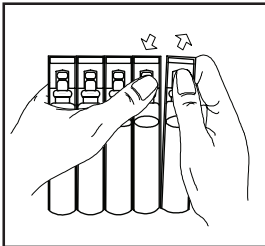
Manufacturer/Product Registration Holder:

AIN MEDICARE SDN. BHD.

Lot 4933, 4934 & PT2464, Jalan 6/44, Kaw. Perindustrian Pengkalan Chepa 2, 16100 Kota Bharu, Kelantan, MALAYSIA

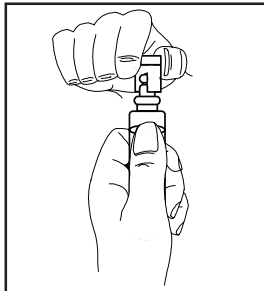
HANDLING INSTRUCTION**AIN MEDICARE INJECTION FLUID IN POLYETHYLENE (PE) AMPOULE**

1)



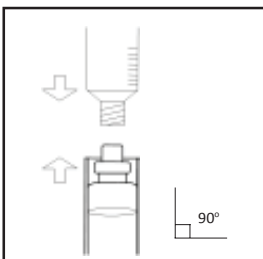
Detach ampoule along parting line

2)



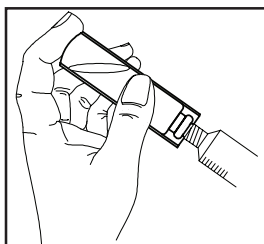
Twist off tab

3)



Attach luer lock syringe, luer slip syringe or hypodermic needle vertically to the ampoule.

4)



Draw out injection fluid using luer lock syringe, luer slip syringe or hypodermic needle. Ensure the connection is tightly secured for the liquid.